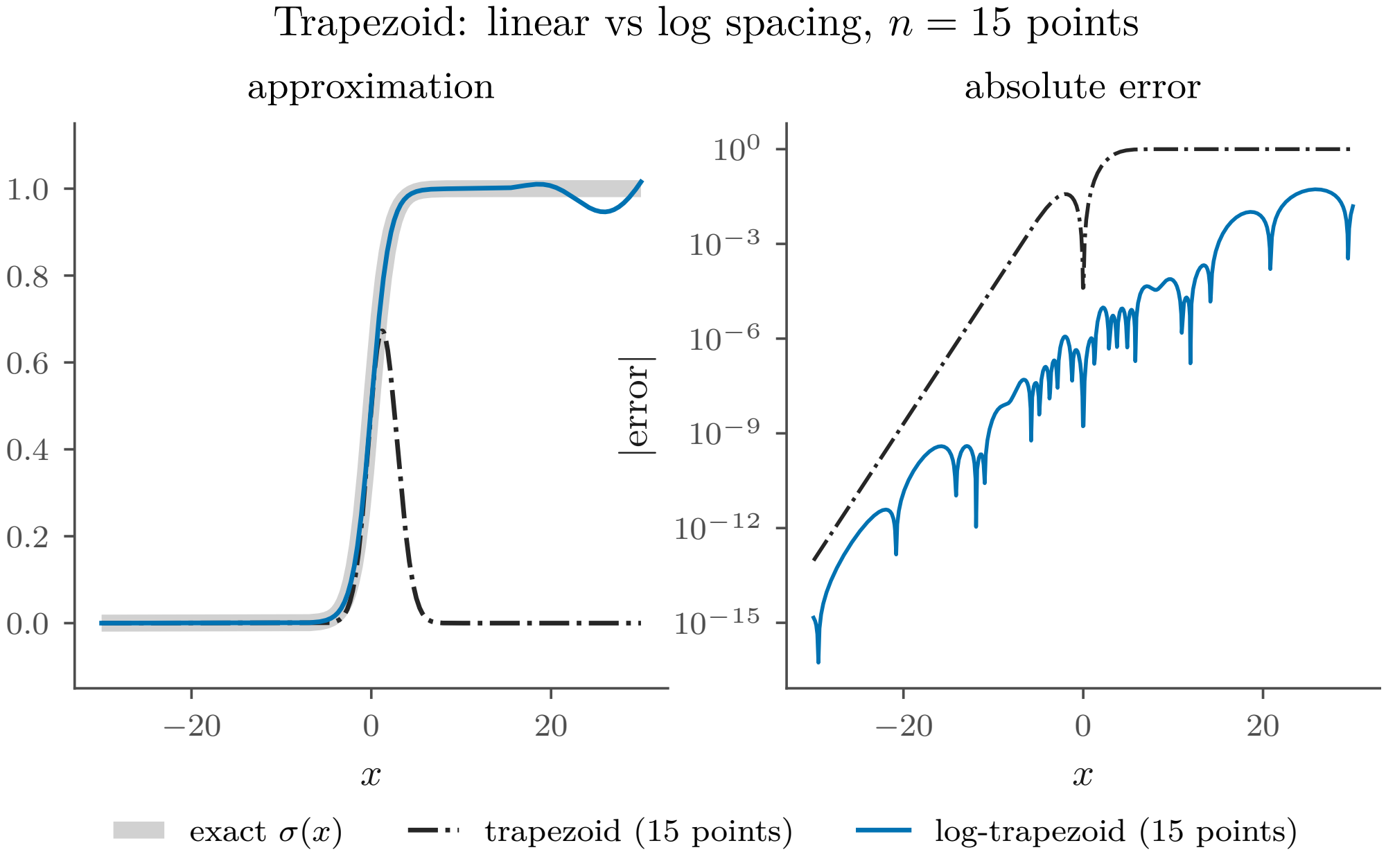

Trapezoid: linear vs log spacing, $n = 15$ points
approximation
absolute error
1.0
0.8
0.6
0.4
0.2
0.0
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
|error|
−20
0
20
x
exact $\sigma(x)$
trapezoid (15 points)
log-trapezoid (15 points)